Lecture 19 (153-248)

1. Prediction recursion
2. How to select p in AR(p)?
3. Uncertainty intervals (Covariance Recursion)
4. Moving Average models (MA)

Prediction Recursion

$$\begin{cases} r_i = 0.7\, r_{i-1}, & i = 1, 2, \ldots \\ r_0 = 2 \end{cases}$$

Solution: $r_i = r_0 (0.7)^i$

or $r_i = 2(0.7)^i$

$$\begin{cases} r_i = 5 + 0.7\, r_{i-1} \\ r_0 = 2 \end{cases} \rightarrow r_i - \frac{5}{0.3} = 0.7\left(r_{i-1} - \frac{5}{0.3}\right)$$

$$\frac{5}{0.3} - 0.7 \times \frac{5}{0.3} = 5$$

$$r_i - \mu = 0.7\,(r_{i-1} - \mu)$$

$$r_i = \mu + 0.7\,(r_{i-1} - \mu)$$

$$= \boxed{\mu\, 0.3} + 0.7\, r_{i-1}$$

$r_0 = 2$ or $r_0 - \mu = 2 - \mu$

$= 2 - \frac{5}{0.3}$

Solution: $r_i = \mu + (0.7)^i (r_0 - \mu)$

$\xrightarrow{i \to \infty}$ μ (does not depend on r_0)

Second-Order Recursion

$$r_i = 0.7\, r_{i-1} + 0.1\, r_{i-2}$$

$r_0 = 3, \quad r_1 = 10$

What is the formula for r_i?

Polynomial corresponding to the recursion:

$$1 - 0.7z - 0.1z^2$$

$$r_i = \phi_1 r_{i-1} + \phi_2 r_{i-2} \longrightarrow \boxed{1 - \phi_1 z - \phi_2 z^2}$$

Find roots of the polynomial: z_1, z_2

$z_1^{-i} \longrightarrow$ Always solves the recursion.

$$r_i = \phi_1 r_{i-1} + \phi_2 r_{i-2}$$

$r_i = z_1^{-i}$

$$z_1^{-i} = \phi_1 z_1^{-(i-1)} + \phi_2 z_1^{-(i-2)}$$

$$z_1^{-i}\left[\underbrace{\phi_1 z_1 + \phi_2 z_1^2}_{=1}\right] = z_1^{-1}$$

Next observation:

$r_i = c_1 z_1^{-i} + c_2 z_2^{-i}$ also solves the recursion for every c_1, c_2.

$r_0 = 3, \quad r_1 = 10$

$c_1 z_1^{-0} + c_2 z_2^{-0} = 3, \quad c_1 z_1^{-1} + c_2 z_2^{-1} = 10$

Solve for c_1, c_2

Solution: $c_1 z_1^{-i} + c_2 z_2^{-i}$

③ $r_i = \phi_1 r_{i-1} + \phi_2 r_{i-2} + \cdots + \phi_p r_{i-p}$

$r_0, \ldots, r_{p-1}$

$1 - \phi_1 z - \phi_2 z^2 - \cdots - \phi_p z^p$

Roots: $z_1, \ldots, z_p$

④ $r_i = \phi_0 + \phi_1 r_{i-1} + \phi_2 r_{i-2} + \cdots + \phi_p r_{i-p}$

$\Updownarrow$

$r_i - \mu = \phi_1 (r_{i-1} - \mu) + \cdots + \phi_p (r_{i-p} - \mu)$

$\mu = \dfrac{\phi_0}{1 - \phi_1 - \cdots - \phi_p}$

$r_i = \mu + c_1 z_1^{-i} + c_2 z_2^{-i} + \cdots + c_p z_p^{-i}$

for some $c_1 \cdots c_p$

Suppose all $|z_j| > 1$, then $r_i \longrightarrow \mu$

STABLE

does not depend on the initial conditions.

NON-STABLE: $|z_j| < 1$ for some j

Connection to AR(p)

$\hat{\phi}_0, \hat{\phi}_1 \ldots \hat{\phi}_p$

$$\hat{y}_{n+i} = \hat{\phi}_0 + \hat{\phi}_1 \hat{y}_{n+i-1} + \cdots + \hat{\phi}_p \hat{y}_{n+i-p}$$

$$\rightarrow \hat{y}_n = y_n, \hat{y}_{n-1} = y_{n-1} \cdots, \hat{y}_{n+1-p} = y_{n+1-p}$$

AR polynomial: $1 - \phi_1 z - \phi_2 z^2 - \cdots - \phi_p z^p$.

degree p

does not depend on ϕ_0.

$z_1 \ldots z_p$

$|z_j| > 1$ for all j $\rightarrow$ STABLE $\longrightarrow$ predictions will converge to $\frac{\hat{\phi}_0}{1 - \hat{\phi}_1 - \hat{\phi}_2 \cdots - \hat{\phi}_p}$

STABLE AR(p)
CAUSAL STATIONARY AR(p)

AutoReg

Determination of p for AR(p)

$$y_t = \phi_0 + \phi_1 y_{t-1} + \cdots + \boxed{\phi_p} y_{t-p} + \varepsilon_t$$

Fit AR(p) & look at estimated coefficient of ϕ_p

$\rightarrow \hat{\phi}_{pp}$

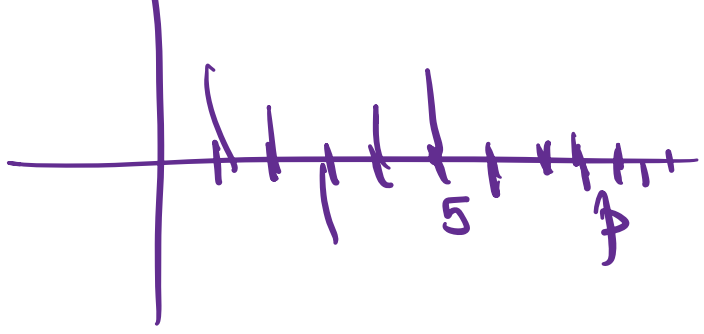
AR(1) $\rightarrow \hat{\phi}_{11}$

AR(2) $\rightarrow \hat{\phi}_{22}$

AR(3) → $\hat{\phi}_{33}$

⋮

AR(p) → $\hat{\phi}_{pp}$



AR(5)

$$\phi_0 + \phi_1 y_{t-1} + \cdots + \phi_5 y_{t-5}$$

$\phi_6 y_{t-6} + \cdots + \phi_{10} y_{t-10}$

Prediction Uncertainties

$$\text{var}\left(y_{n+i} \;\middle|\; \begin{matrix} y_1 \cdots y_n \\ \text{\& parameters} \end{matrix} \right) = V_i$$

How to calculate V_i for $i = 1, 2 \ldots$?

$$\text{Cov}\left(\begin{matrix} y_{n+1} \\ \vdots \\ y_{n+k} \end{matrix} \;\middle|\; \begin{matrix} y_1 \cdots y_n \\ \text{\& parameters} \end{matrix} \right) = \Gamma_k$$

$$\text{Cov}\begin{pmatrix} X_1 \\ \vdots \\ X_k \end{pmatrix} = \left[\text{Cov}\begin{pmatrix} X_1 \\ \vdots \\ X_{k-1} \end{pmatrix} \;\; \text{Cov}\left(\begin{matrix} X_1 \\ \vdots \\ X_{k-1} \end{matrix}, X_k \right) \right]$$

$\text{Cov}(X_k, (X_1 \cdots X_{k-1}))$ $\quad \text{var}(X_k)$

$\text{Cov}(X_1, X_k)$ is the (i,j)th entry in $\text{Cov}(X_i, X_j)$

$k \times k$

$1 \times (k-1)$ $\left(\gamma_{k-1} \right)$

Γ_k in terms of Γ_{k-1}

Γ_k

$$\text{Cov}\left(\begin{pmatrix} y_{n+1} \\ \vdots \\ y_{n+k} \end{pmatrix} \middle| \begin{matrix} y_1 \dots y_n \\ \text{all parameters} \end{matrix} \right)$$

$$\text{Cov}\left(\begin{pmatrix} y_{n+1} \\ \vdots \\ y_{n+k-1} \end{pmatrix} \middle| \begin{matrix} y_1 \dots y_n \\ \text{all parameters} \end{matrix} \right)$$

$$\Gamma_k = \begin{bmatrix} & & \\ & \Gamma_{k-1} & \\ & & \end{bmatrix}_{k \times k}$$

$$\text{Cov}\left(\begin{pmatrix} y_{n+1} \\ \vdots \\ y_{n+k-1} \end{pmatrix}, \; y_{n+k} \middle| \dots \right)$$

$$= \text{Cov}\left(\begin{pmatrix} y_{n+1} \\ \vdots \\ y_{n+k-1} \end{pmatrix}, \; \phi_0 + \phi_1 y_{n+k-1} + \phi_2 y_{n+k-2} + \dots + \phi_p y_{n+k-p} + \varepsilon_{n+k} \middle| \dots \right)$$

$$= \mathrm{Cov}\left(\begin{pmatrix} y_{n+1} \\ \vdots \\ y_{n+k-1} \end{pmatrix}, \phi_1 y_{n+k-1} + \cdots + \phi_p y_{n+k-1} \cdots \right)$$

$$= \mathrm{Cov}\left(\begin{pmatrix} y_{n+1} \\ \vdots \\ y_{n+k-1} \end{pmatrix}, a^T \begin{pmatrix} y_{n+1} \\ \vdots \\ y_{n+k-1} \end{pmatrix} \cdots \right)$$

for some a

$$= \Gamma_{k-1}\, a$$

$$\Gamma_k = \begin{pmatrix} \Gamma_{k-1} & \Gamma_{k-1} a \\ a^T \Gamma_{k-1} & a^T \Gamma_{k-1} a \end{pmatrix} \qquad k = 1, 2 \cdots$$

$$\Gamma_1 = \mathrm{Cov}\left(y_{n+1} \mid \underset{\text{parameters}}{y_1 \cdots y_n} \right)$$

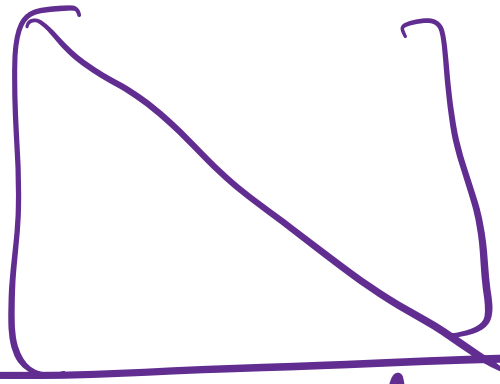
$$= \mathrm{var}\left(y_{n+1} \mid \underset{\text{parameters}}{y_1 \cdots y_n} \right)$$

$$= \mathrm{var}\left(\varepsilon_{n+1} \mid \underset{\text{parameters}}{y_1 \cdots y_n} \right) = \sigma^2$$

$\Gamma_1 = \sigma^2$

$$\Gamma_2 = \begin{pmatrix} \sigma^2 & * \\ * & * \end{pmatrix}$$

$\Gamma_3 = \begin{pmatrix} \Gamma_2 & * \\ * & * \end{pmatrix}$, $\Gamma_4 \cdots$

$\Gamma_k =$ 

MA (Moving Average)

MA(q) for some $q \geq 1$

$$y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \cdots + \theta_q \varepsilon_{t-q}$$

where $\varepsilon_t \overset{iid}{\sim} N(0, \sigma^2)$ & $\theta_1, \ldots \theta_q, \mu, \sigma$ are unknown parameters

Special cases

MA(0) : $\overset{iid}{\sim} N(\mu, \sigma^2)$

MA(1) : $y_t = \mu + \varepsilon_t + \theta \varepsilon_{t-1}$

$E y_t = \mu$, $\text{var}(y_t) = (1 + \theta^2)\sigma^2$

$$\text{Cov}(y_t, y_{t+1}) = \text{Cov}(\varepsilon_t + \theta \varepsilon_{t-1}, \varepsilon_{t+1} + \theta \varepsilon_t)$$

$$= \text{Cov}(\varepsilon_t, \theta \varepsilon_t)$$

$$= \theta \sigma^2$$

$$\Rightarrow \text{Corr}(y_t, y_{t+1}) = \frac{\theta\sigma^2}{\sqrt{\text{var}(y_t)\ \text{var}(y_{t+1})}}$$

$$= \frac{\theta}{1+\theta^2}$$

$$\text{Cov}(y_t, y_{t+2}) = 0$$

More generally, MA(q) has correlations upto lag q:

$$\text{Corr}(y_t, y_{t+h}) = \begin{cases} \text{may be non zero} & |h| = 1, \dots, q \\ 0 & \text{for } |h| > q \end{cases}$$

$$y_t = \sum \psi_j \varepsilon_{t-j}$$

$$\text{Cov}(y_t, y_{t+h})$$

$$= \sum \psi_j \psi_{j+h}$$

Parameter Estimation in MA(q)

ARIMA (p, d, q)

p: AR, d: differencing, q: MA

$\{y_t\}$: MA(q) : ARIMA(0, 0, q)

MA(1)

$$y_t = \mu + \varepsilon_t + \theta \varepsilon_{t-1}, \quad \varepsilon_t \overset{iid}{\sim} N(0, \sigma^2)$$

Likelihood μ, θ, σ^2

$$\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \sim N\left(\begin{pmatrix} \mu \\ \vdots \\ \mu \end{pmatrix}, \begin{bmatrix} \sigma^2(1+\theta^2) & \theta\sigma^2 & & 0 \\ \theta\sigma^2 & \sigma^2(1+\theta^2) & \theta\sigma^2 & \\ & \theta\sigma^2 & \ddots & \theta\sigma^2 \\ 0 & & \theta\sigma^2 & \sigma^2(1+\theta^2) \end{bmatrix} \right)$$

$$\mathrm{var}(y_t) = (1+\theta^2)\sigma^2$$

$$\mathrm{Cov}(y_t, y_{t+1}) = \theta\sigma^2$$

$\sigma^2(1+\theta^2)$

$$\Sigma = \begin{bmatrix} \sigma^2(1+\theta^2) & \theta\sigma^2 & & 0 \\ \theta\sigma^2 & \sigma^2(1+\theta^2) & \theta\sigma^2 & \\ & \theta\sigma^2 & \ddots & \theta\sigma^2 \\ 0 & & \theta\sigma^2 & \sigma^2(1+\theta^2) \end{bmatrix}$$

COVARIANCE MODEL

$$\frac{1}{\sqrt{\det \Sigma}} \exp\left(- \frac{(y - \text{mean})^T \Sigma^{-1} (y - \text{mean})}{2} \right)$$

KALMAN FILTER

$$
\left.
\begin{aligned}
y_1 &= \mu + \varepsilon_1 + \theta\,\varepsilon_0 \\
y_2 &= \mu + \varepsilon_2 + \theta\,\varepsilon_1 \\
&\vdots \\
y_n &= \mu + \varepsilon_n + \theta\,\varepsilon_{n-1}
\end{aligned}
\right\}
$$

Trick: Take $\varepsilon_0 = 0$. Then

$$
\begin{aligned}
\varepsilon_1 &= \hat{\varepsilon}_1 = y_1 - \mu \\
\varepsilon_2 &= y_2 - \mu - \theta(\varepsilon_1) \\
\varepsilon_3 &= y_3 - \mu - \theta\,\varepsilon_2 \\
&\vdots \\
\varepsilon_n &= y_n - \mu - \theta\,\varepsilon_{n-1}
\end{aligned}
$$

$$
\left\{
\begin{aligned}
\hat{\varepsilon}_1 &= y_1 - \mu \\
\hat{\varepsilon}_2 &= (y_2 - \mu) - \theta\,(y_1 - \mu) \\
\hat{\varepsilon}_3 &= (y_3 - \mu) - \theta\,[(y_2 - \mu) - \theta\,(y_1 - \mu)] \\
&\vdots \\
\hat{\varepsilon}_n &=
\end{aligned}
\right.
$$

→ This gives $(\varepsilon_1 \dots \varepsilon_n)$ in term of $y_1 \dots y_n \& \mu, \theta$

Likelihood:

$$\prod_{i=1}^{n} \frac{1}{\sigma} \exp\left[- \frac{\hat{\varepsilon}_i^2}{2\sigma^2} \right]$$

$$= \sigma^{-n} \exp\left[- \frac{S(\mu, \theta)}{2\sigma^2} \right]$$

$$S(\mu, \theta) = \sum_{i=1}^{n} \hat{\varepsilon}_i^2$$

$$(\hat{\mu}, \hat{\theta}) = \arg\min S(\mu, \theta)$$

non linear function of μ, θ

Bayesian posterior:

$$\left[\frac{1}{S(\mu, \theta)} \right]^{n/2}$$

$$= \left[\frac{1}{S(\hat{\mu}, \hat{\theta}) + \text{square term}} \right]^{n/2}$$

ARIMA

MA(q)

~~ARMA~~
ARIMA
SARIMA

